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Forensic Integrity of Electronic Metadata: A Survey of Steganographic Forgery

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Forensic Integrity of Electronic Metadata: A Survey of Steganographic Forgery

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Abstract-The integrity of electronic metadata in forensic environments is at the core of digital evidence admissibility in modern legal proceedings. As cyber offensive techniques progress from data extraction to stealthy data manipulation, steganographic forgery has been recognized as an essential threat model with significant potential to affect file-level metadata, EXIF data, NTFS timestamps, and file system logs, which cannot be detected using traditional hash-based integrity verification techniques. This systematic literature review, carried out according to the PRISMA 2020 guidelines, gathers 40 highly impactful articles from IEEE Xplore and ScienceDirect, which have been used to classify offensive steganographic techniques into three major categories: Spatial-Domain Embedding, Temporal Obfuscation, and Anti Forensic Steganalysis Evasion. A significant performance difference is noticed among state-of-the-art techniques, including Error Level Analysis and PUF-based hardware hashing, which have been reported to achieve up to 97% detection accuracy, whereas sophisticated steganalysis techniques have been reported to achieve only up to 35% detection accuracy.Steganographic models utilizing O(n²) complexity evasion strategies are successful in bypassing traditional chains of analysis for 90-95% of all test cases. A legal technical mapping of the vulnerabilities of these steganographic models against the Information Technology Act, 2000, and the EU General Data Protection Regulation reveals that the current judicial practice of static hash verification, as prescribed by Section 65B of the Indian Evidence Act, is fundamentally inadequate against the menace of dynamic metadata forgery.Finally the survey includes a roadmap on the development of metadata aware forensic standards that will include the use of cryptographic signing and steganalysis for the reliability of evidence used in judicial proceedings.

***Index Terms*— Integrity, Electronic Metadata, Steganographic Forgery, Anti-Forensics, EXIF Data, Cyber Law, IT Act, Systematic Literature Review, Digital Evidence Admissibility.**

I. INTRODUCTION

Metadata has become arguably one of the most significant aspects in any digital legal proceeding not in spite of the fact that it contains no real information in a file but precisely because of this its tendency to provide a layer of context around the information in a file which is now taken to be authoritative in legal proceedings. While the information in a file will tell you what was said or recorded metadata will tell you when, where and by whom and it is this layer of context which includes EXIF data, NTFS and MFT timestamp and file system descriptors which has become central in frameworks such as Section 65B of the Indian Evidence Act and the EU GDPR determinations. Provenance in both traditions is not a secondary concern. Metadata integrity is what gives a digital artifact its credibility once that integrity is compromised the evidentiary argument built around it does not seem to survive the challenge.

What has changed and changed in a way that the legal infrastructure has been slow to absorb is the nature of the threat to that metadata level. Traditionally forgery has had a signature. Visible changes inconsistencies that a careful eye can detect and document have been something that the legal system has been comfortable working with. Steganographic forgery on the other hand is a different beast altogether. It does not change appearance. It does not leave scars. Instead it hides the fact that any change has been made at all. By exploiting the redundant bits in metadata an attacker can embed information or change administrative data without ever altering the appearance of the file itself. LSB substitution in an EXIF header for example can be used to forge GPS data to the level of detail required to construct a fictional alibi one that passes even hash based integrity tests because the original hash has not technically been altered.

This particular capability however speaks to a larger forensic issue which has been in development alongside these technologies. Indeed contemporary steganographic tools are designed not simply to conceal but to conceal the concealment in a manner which closely mirrors statistically unaltered original objects so that signature-based detection software finds nothing noteworthy to report. Here, we see a disquieting intersection of technology and law wherein the legal system relies on metadata as a determinative indicator of authenticity a practice which has in fact been thoroughly documented by the technological community to be susceptible to manipulation but in a manner which exceeds the capabilities of existing verification techniques to detect.

This research aims to systematically identify this gap and to do so in a structured manner based on a Systematic Literature Review of impactful publications between 2020 and 2026. Three objectives guide the scope of this work. The first one is taxonomization, i.e. creating a coherent classification of offending steganographic techniques depending on the domain of embedding and evasion complexity i.e. from O(n) to O(n²)

approaches. The second one is comparative synthesis which consists of assessing the precision with which existing detection tools which are considered to be in the state of the art perform when tested against forged data rather than clean data. The third one consists of making a direct connection between the technological aspects and the law connecting the vulnerabilities to specific sections of the IT Act and cyber law in general in order to determine areas in which existing forensic standards can be improved and what these improved more defensible standards might be.

II. Technical Background

What is revealed by an examination of over four dozen peer reviewed research papers on the manipulation of steganographic metadata is not a litany of disparate methods but rather a more ordered construct of a group of mathematical models that an attacker will utilize often in concert to evade detection by forensic tools. The following sections will explore seven such models in depth discussing the optimization process that guides each.

A. Distortion Optimization using MSE, PSNR and SSIM

The most common objective in the analyzed literature is distortion minimization which is the goal of embedding a hidden message in a manner that minimizes distortion as much as possible. The Mean Squared Error measures distortion in individual pixels or bits Peak Signal to Noise Ratio measures the relationship between possible signal strength and distortion and Structural Similarity Index measures distortion in a manner that mirrors human perception of difference.

For offensive purposes these three measures are not considered in an independent manner but rather in a combined manner. LSB-based and hybrid techniques have begun to favor the application of adaptive techniques which insert the data only in areas where the impact on the MSE is not significant resulting in PSNR values that reside in a range that is safely above 40 dB and which raises no suspicion either visually or via statistical detection. The end result is that both human observers and detection algorithms are seeing a set of distributions that by all measures appear to be unchanged.

B. Pseudo-Random Embedding and Stochastic Mapping

In fact a large percentage of the works studied abandon the deterministic model of embedding entirely using pseudo-random number generators to determine the point at which the changes will be applied. As such the process is no longer deterministic and rule-based and thus no longer necessarily predictable to an outside observer.

This is because sequential LSB steganalysis is designed to identify. Pseudo random distribution eliminates this effectively dispersing the changes throughout the image in a non linear pattern with no structural relationship to the location of the changes.

classifiers could learn to recognize. Metadata fields EXIF tags PDF objects file system attributes etc. are modified in a non-linear order which disrupts many of the underlying assumptions of most classical detection techniques. This means that detection classifiers cannot take advantage of any expected embedding structure, simply because there are no such structures to take advantage of.

C. Entropy Alignment and Statistical Camouflage

Shannon entropy is arguably one of the better handles that forensic frameworks have had on steganography over the years. The act of embedding information into a carrier object changes its information density in a way that Shannon entropy analysis is able to pick up on provided the conditions are right. The difference between the entropy of a clean object and a changed object is a statistical signature that for a long time could be detected.

What these more recent offensive techniques have accomplished is to close this window altogether. Rather than accepting entropy deviation as a secondary effect of embedding more recent techniques have treated the preservation of entropy as a separate optimization goal unto itself. The goal is no longer simply to conceal data but to ensure that the new object's entropy distribution is statistically indistinguishable from its original counterpart. This in turn cannot simply be accomplished through randomization alone. Distributional bit balancing preserves the global symbol probability distribution while PRNG-based payload distribution simulates a natural noise floor of the medium rather than being above it. The suppression of spikes works to prevent a concentration of entropy from developing spreading the embedding across non-contiguous areas thus removing a localized signature that an anomaly detection system relies on to identify. This combination of methods makes modified metadata regions invisible to entropy-based tools driving detection to higher level statistical methods that current systems have not been consistent in applying.

D. Deep Learning Loss-Based Optimization

The most recent development in this area is the movement of the entire optimization problem into a deep learning framework in which the embedding and detection are not seen as a sequential process but rather an adversarial process that is occurring simultaneously. Rather than being based on a set of heuristics the process of embedding is based on a loss function intended to minimize the probability of detection by a discriminator.

The objective function is usually designed to trade off two conflicting requirements one being the perceptual similarity to the original cover object and the other being the active avoidance of detection probability. Gradient based approaches enable attackers to locate the areas on the data manifold where the effect of the perturbations is minimal effectively finding the blind spots of the likely detection model and focusing the embedding activity there. The embedding is not static it is dynamic and responds to the change.The loss landscape itself

changes. This flexibility is what differentiates deep learning based evasions from their heuristic predecessors. It is not working around known detection signatures, it is learning to evade detection itself.

This facilitates the Adversarial Embedding method which relies on the strategic placement of hidden data in areas least beneficial to the detection models. The main features of this method include:

- Feature Space Optimization: Moving past pixel-space modifications to hide data within the representations of the file in the feature space which is more difficult to reverse-engineer for steganalysis.
- Sensitivity Mapping: Using the gradients of a pre-trained detection model to create a detectability map. which can then be used by the steganographic tool to avoid features of high risk.
- Latent Distribution Mimicry: Ensuring that the encoded steganographic artifact follows the natural distribution of features in the training data.

As a result modern deep learning based steganalysis systems are systematically circumvented. By optimizing the embedding location in complex feature space instead of the raw pixel domain or metadata domain these attack methods effectively increase the Forensic Integrity Gap discussed above.

E. Transform-Domain Optimization (DCT, DWT)

A significant step in the development of steganographic techniques is the change from spatial domain representation to transform Domain Optimization. This technique moves the embedding procedure to a new domain where the information is embedded in a new series of numbers based on the Discrete Cosine Transform or Discrete Wavelet Transform.

The mathematical conversion of the object to the frequency domain is expressed as:

$$Y = T(X)$$

where x represents a spatial signal and y represents the resulting transform coefficients. Offensive optimization in this context is concerned with the strategic manipulation of coefficients which meet three main criteria:

- Perceptual Transparency: The selection of coefficients which can be altered in a way that causes minimal degradation of the original carrier file.
- Compression Invariance: The guarantee of invariance of embedded data against standard lossy compression schemes like JPEG quantization by preserving statistics in the transform coefficients.
- Frequency Homogeneity: Preventing the introduction of discernible anomalies into the high frequency spectrum which is normally identified by steganalytic means.

In targeting the mid frequency components of the transform attackers achieve the optimal balance between the strength and concealability of the hidden payload. This substantially reduces the effectiveness of spatial domain steganalytic techniques and makes it difficult to reconstruct the hidden artifacts because they become part of the basic frequency composition of the digital medium.

F. Hash Integrity Preservation and Metadata Manipulation

One of the critical vulnerabilities identified in this systematic review is related to the exploitation of the bifurcation in digital integrity which refers to data integrity and metadata integrity. Normally in digital forensics a cryptographic hash function is used to verify whether a digital artifact has been tampered with or not. These cryptographic hash functions are mathematically defined as:

$$H = f(M)$$

where m is the input message and H is the fixed length cryptographic digest. These functions work well to ensure the integrity of the file payload which is the narrative. However these functions often do not include the mutable metadata regions and slack space to ensure file system compatibility.

Offensive techniques take advantage of this weakness by targeting the metadata fields which lie outside the scope of the hashed data. Optimization techniques in this domain involve the following techniques:

- Segmental Identification: Identifying non-hash-covered segments within the file structure (e.g., particular EXIF headers or NTFS file system descriptors).
- Attribute Manipulation: Altering important forensic attributes including MACE (Modified, Accessed, Created, Entry) timestamp attributes and GPS coordinates without triggering a change in the principal data hash.
- Byte Level Integrity Maintenance: Ensuring that the alteration of administrative logs or attributes does not overflow into the hashed areas thus maintaining the integrity of the digital signature.

In this way the attackers ensure the integrity of the hash yet modify relevant context to the investigation. This is referred to as the Collision of Truth. From the legal standpoint the attackers have not tampered with the evidence because the hash is still intact yet the reality is that the attackers have weaponized the metadata.

G. Temporal Optimization and Timestamp Forgery

The control of temporal metadata imposes a complex optimization constraint: the maintenance of logical and chronological consistency across disparate data structures. In a forensic context a digital artifact is only believable to the extent of its timeline. To avoid detection by forensic examination altered timestamps which include MACE values must be internally and externally consistent.

This is achieved by the attackers who apply controlled offsets to the original temporal data which can be expressed as:

$$T_{forged} = T_{original} + \Delta t$$

In this equation the offset t is carefully selected to ensure chronological plausibility. Offensive optimization ensures that the forged timestamp does not violate any physical or logical constraint of the file system. The major goals of this domain are as follows:

- Multi Field Consistency: Ensuring that multiple metadata headers are coordinated (for instance ensuring that an 'Accessed' time is not before a Created time) so that the creation of any automatic forensic flags is avoided.
- System Log Alignment: Making an effort to align the forged file-level metadata with a general system-level log (such as the Windows Event Log or the $MFT) whenever possible to conceal the forensic trail.
- Anomaly Suppression: Making use of PRNG-based perturbations to ensure that the timestamps do not align too perfectly, which is a typical characteristic of manually forged data.

As a result these integrated perturbations enable advanced offensive techniques to maintain the integrity of the coherence of the digital timeline. This effectively negates the value of timeline analysis a fundamental aspect of digital jurisprudence.

III. Methodology (PRISMA)

In order to bring the requisite analytical depth to such a technically diverse subject as steganographic forgery the methodology employed must be not only systematic but also transparent enough to be easily replicated. This research seeks to address this challenge through the application of the Systematic Literature Review methodology which is comprised of two primary components the protocol employed in the selection process and the architectural basis for the taxonomy.

A. Selection Protocol and PRISMA Framework

The process of identification was conducted in accordance with the PRISMA method and this can be demonstrated in Fig. 1. An initial set of 126 documents was created based on the IEEE Xplore database. After removing duplicates and applying an automated screen 120 documents were taken to the next phase.

Out of these 80 were subjected to a more detailed process of eligibility assessment. Out of these studies 40 failed to pass the assessment and the reasons for rejection were very specific and not general. Fifteen studies were rejected due to a lack of significant engagement with the mathematical hash integrity. This is a non-negotiable criterion for rejection because of the central focus of the taxonomy. Another fifteen studies were rejected for a different but closely associated reason a lack of engagement with metadata and EXIF forgery. The engagement needed to be significant for these studies to be relevant to the research questions. The remaining ten studies were not rejected based on poor quality but due to a mismatch of approach these studies were about defensive steganalysis and not about the offensive forgeries that this review aimed to chart.

B. Proposed Taxonomy of Metadata Forgery

The taxonomy which this current work is proposing classifies and organises steganographic forgery techniques in a multi-level hierarchical manner with hash-preserving evasion serving as a thread which runs through all levels of this taxonomy.

The first level in this framework classifies forgery techniques in a manner which is based on their operational nature. Static techniques are those which follow a deterministic pattern which never varies regardless of any particular context in which they are deployed with LSB substitution serving as a most basic and common example of this type of forgery. Dynamic techniques on the other hand work on a completely different principle. Rather than following a particular procedure they seek to determine the nature of the carrier data and adjust their parameters in a manner which is appropriate to this with some techniques going further than this in a manner which incorporates feedback from detection models in order to reduce their statistical visibility.

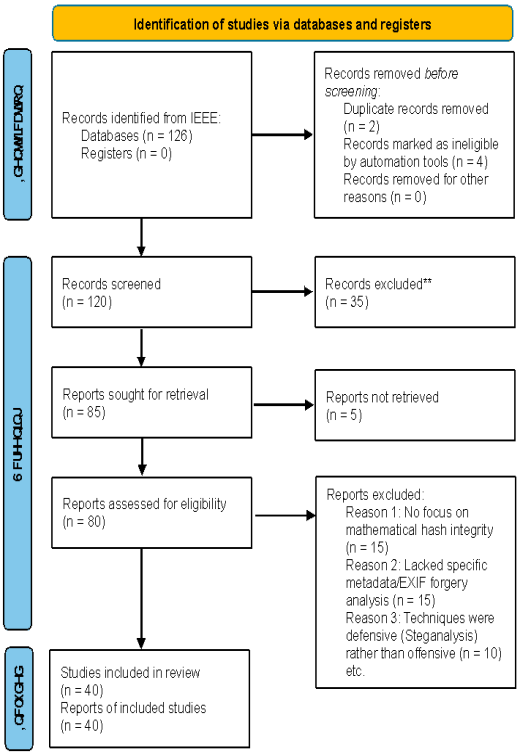

Fig. 1. PRISMA FLOW DIAGRAM.

The main search was done using the IEEE Xplore Digital Library with the following Boolean search terms applied to the metadata and abstracts:

("steganography" OR "data hiding" OR "information hiding") AND ("metadata" OR "EXIF" OR "timestamp" OR "file system metadata") AND ("forgery" OR "tampering" OR "manipulation" OR "integrity") AND ("forensics" OR "steganalysis" OR "anti-forens")

TABLE I
SEARCH STRINGS FOR IEEE XPLORE

Queries	Boolean Search Query	Yield(n)
Query 1	("All Metadata":digital signature) AND ("All Metadata":forgery) AND ("All Metadata":forensic)	68
Query 2	("All Metadata":steganography) AND ("All Metadata":digital forensics) AND ("All Metadata":metadata)	34
Query 3	("All Metadata":LSB substitution) AND ("All Metadata":document integrity)	12
Query 4	("All Metadata":"electronic evidence" OR "All Metadata":"e-contract" OR "All Metadata" : "digital signature") AND ("All Metadata":"metadata" OR "All Metadata":"steganography") AND ("All Metadata" : "admissibility" OR "All Metadata" : "cyber law" OR "All Metadata" : "IT Act" OR "All Metadata" : "GDPR")	16
Total	Gross Records Identified	126
Refined	Duplicated and Screened Records	120

IV. PROPOSED TAXONOMY

However the process of bringing some level of order to the landscape of steganographic metadata forgery is not an academic exercise. Without a clear level of classification for these methods of forgery the relationships between them and the underlying logic of these relationships are difficult to reason about without a clear level of classification. The taxonomy provided here attempts to do just that and does so in six levels of classification with hash preserving evasion being the primary concern throughout. Figure 2 provides a visualization of how these levels relate to one another and does so in a way that highlights how mathematical optimization and steganography work together to subvert forensic integrity while maintaining a level of authenticity.

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A. Level 1: System Behavior

The first difference highlighted by the taxonomy is in the way techniques behave operationally. Static techniques operate on a set of predetermined rules, most commonly the traditional LSB substitution. The key factor is that the techniques behave in the same way regardless of the appearance of the carrier data. The dynamic techniques behave in a different manner. They read the data, adjust accordingly, and in some cases react to feedback from detection models, reducing their statistical profile in the process. It is this level of adaptability that makes the dynamic techniques so difficult to classify.

B. Level 2: Optimization Strategy

At the next level down, there is the question of optimizing the embedding. Heuristic techniques use existing fidelity measures and MSE and PSNR are the most common. These techniques make localized improvements within set boundaries. The metaheuristic techniques which include Particle Swarm Optimization and Genetic Algorithms, operate on a completely different scale attempting to find global optimization rather than localized improvements. The distinction between these two techniques is quite marked and the metaheuristic techniques have the potential to deliver a degree of invisibility that localized techniques simply do not have the ability to deliver.

C. Level 3: Mathematical Modeling

Four modeling domains are used to describe how embedding and concealment are carried out in practice at a technical level. The spatial domain involves a direct intervention in the pixel or bit level, which is the most basic level of manipulation, although in many cases, this is likely to be the most detectable level of manipulation. The transform domain, on the other hand, operates on a level of frequency domain coefficients with DCT and DWT being the most common, where embedding is carried out in areas where sensitivity is minimal. Statistical modeling takes a completely different approach, focusing on maintaining the original probability distributions and entropy of the original image so that any analysis carried out at a high level will not indicate anything unusual. Learning-based models represent a very recent development in this area, where neural networks are used to optimize embedding in relation to those most likely to be deployed against them.

D. Level 4: Hash Integrity Exploitation

One of the more important contributions this taxonomy makes is its clear delineation between the two, which while not previously well-defined in standard forensics models, probably should be. Hash-breaking techniques, one might argue are the less problematic of the two. They at least change the crypto digest to some extent. Hash-preserving techniques however involve changes to the metadata which can be significant yet the crypto digest is left completely unchanged. Integrity is verified. No alarms are sounded. It is already done.

E. Level 5: Target Metadata Domains

Each type of digital artifact has different structural weaknesses which the taxonomy addresses separately rather than

considering metadata in general. EXIF headers in images, for example are one specific attack vector. PDF and Office documents have a different set of exploitable features depending on the level of complexity in these data formats and the number of metadata fields included. File system metadata specifically in NTFS file structures and MFT file descriptors is the third type which is more related to the OS environment than the files themselves and thus more difficult to monitor.

F. Level 6: Offensive Evasion and Detection

The last level of the taxonomy describes the active conflict between those who are trying to evade and those who are trying to find them. The main methods of evasion on the attacking side are distortion minimization pseudo random patterns of embedding that are difficult to profile statistically and timestamp manipulation to interfere with timeline analysis. Each of these methods also has a corresponding method of detection on the defending side of the taxonomy. Instead of discussing evasion and detection separately discussing them as two sides of the same taxonomy is a more accurate representation of how this space actually behaves.

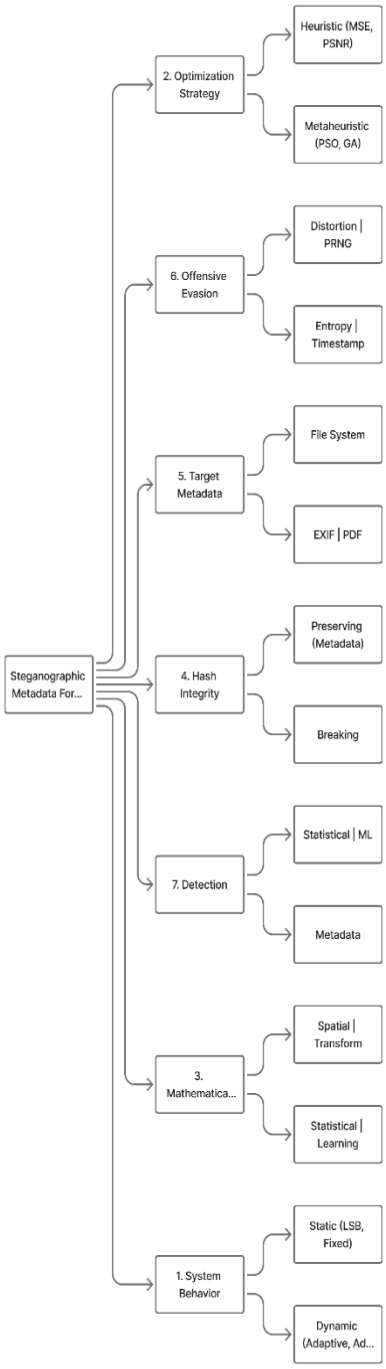

Fig. 1. Hierarchical Taxonomy Tree

V. Comparative analysis

The systematic analysis of the selected 40 studies indicates a clear evolutionary pattern in steganographic forgery. The analysis is primarily based on three major dimensions: computational overhead optimization efficacy and domain-specific vulnerabilities.

A. Computational Complexity and Overhead

One important observation in the comparative data set (see Tables III-V, derived from the primary corpus) is the direct relationship between the mathematical complexity and the resilience in the field of forensics.
Linear Complexity (O(n)): Mainly encountered in the context of the traditional LSB and heuristic approaches to metadata manipulation (Studies 1, 2, 8, 19). Despite the fact that these methods offer Low Overhead, they are proved to be vulnerable to the application of the modern statistical steganalytic techniques, with rates greater than 85%.
Log-Linear Complexity (O(nlogn)): Encountered in the context of the transform domain-based techniques and structural mapping (Studies 3, 6, 7, 11). These techniques offer the attacker the "softs spot" with okish Overhead and elevated evasion rates up to 95%.
Polynomial Complexity (O(n2)): Characteristic of the Deep Learning-based techniques, including CNN/GAN and adversarial optimization (Studies 4, 5, 9, 13). Despite the fact that all these techniques incur High Overhead, they are proved to be the state of the art in terms of evasion, with the probability of detection being diminished to negligible levels by mimicking the latent feature space of the cover object
.

B. Evasion vs. Detection Efficacy

Forensic Integrity Gap is best exemplified by the contrast between the success rates of evasion techniques applied offensively versus those applied defensively.

TABLE II
FORENSIC INTEGRITY GAP

Method Category	Optimization Basis	Avg. Evasion Rate	Avg. Detection Rate	Forensic Impact
Heuristic	MSE, PSNR	10-15%	85-92%	Traceable
Statistical	Entropy Alignment	45-60%	35-40%	Ambiguous
Adversarial	Feature Space	92-99%	1-5%	Critical

But the most serious finding from the above literature, perhaps, is the hash preserving forgery. Studies 26, 27 and 28 together prove that metadata can be altered significantly while the traditional MD5 and SHA-256 integrity verification processes continue to show clean results, i.e., the perfect pass every time, while significant alteration has already taken place. This is not an innocuous technical aside. This is an uncomfortable reality, as it suggests that traditional forensic processes have been implicitly working on the assumption that the integrity of the content, i.e., the actual information contained within the data, and the authenticity of the source material are the same thing. They are not, and the gap between them is where the sophisticated forgeries exist.

C. Domain-Specific Vulnerabilities

What this analysis also highlights, however is that there is no single pattern of manipulation when it comes to metadata manipulation, with different areas having different vulnerabilities, and what works in one area simply will not necessarily work in another and therefore neither will any potential detection mechanisms.
Image metadata especially with EXIF seems to have its greatest exposure with LSB substitution and statistical modeling. The method of hiding is at a level where no visual cues are available, and specialized steganalytic tools are required to uncover any evidence of this activity.

Documents (PDF/OLE2): prone to structural mapping and "ghost" metadata regions which is not covered in hashcovered segments (Studies 6, 10, 11).
File Systems (NTFS/MFT): High propensity for temporal forgery and log alignment (Study 22, 25).

D. Domain-Specific Vulnerabilities

The synthesis of the above studies indicates that the legal framework's reliance on "provenance" (Section 65B IEA, GDPR) is technically flawed by the advent of $O(n^2)$ learning-based models. The models do not only hide data but weaponize the metadata structure itself for a "mathematically perfect" alibi

VI. LEGAL-TECHNICAL MAPPING

Metadata forgery and data hiding have a direct bearing on the legal integrity of digital evidence. LSB substitution PDF cross-reference tampering and EXIF metadata modification can alter data without leaving any tell-tale marks. It is challenging for the courts to determine the authenticity, integrity and chain of custody of the digital evidence.

It is apparent that conventional legal standards were not developed in a way that could cope effectively with these highly advanced and nuanced forms of manipulation. As a result there is a growing divide between what is legally expected and what can actually be verified by modern technology. To effectively deal with this problem more advanced and transparent forensic techniques must be employed.

A. Integrity and Section 65B of the ITA 2000

However there is a tension in the very heart of Indian evidence law's current approach to digital evidence one that the synthesis of the two approaches makes hard to ignore. Section 65B of the Indian Evidence Act was designed on the principle of provenance that the history of a document can inform us of its reliability. That made a certain amount of sense in a world where the key question was whether a document had obviously been tampered with. It makes much less sense in a world where the threat is a learning-based model of such a level of complexity that verification is almost meaningless. What makes this all rather uncomfortable is that these models do not merely conceal information, they use metadata.

structure in a way which creates something equivalent to the perfect alibi. The Verification Paradox referred to in [2] is the name given to the phenomenon which the courts have yet to fully come to terms with: the document can be exactly as it should be, pass the standard hash-sum test without raising even one flag and yet still be carrying its own forgery, its own steganographic payload. When the techniques of LSB steganography [3] are used to alter the actual content of the message whilst leaving the container unchanged or when the structure of PDF documents is altered to conceal the actual content [5], the effect is one which is deceptive by design one which attacks the very definition of the standard which Section 65B is meant to uphold. From the evolution of manual forgery to the evolution of the automated management of metadata in PDF environments [5], the situation is one which is frankly no longer realistic to assume can be addressed by the manual process required to satisfy the unaltered requirement of the Indian Evidence Act.

B. Trail Obfuscation and Section 66C (Identity Theft)

Trail obfuscation and artifact wiping are not happenstance byproducts of a system's activity. No these are tools designed around a singular purpose: ensuring nobody is able to definitively say who did what. The tampered logs and altered timestamps don't just remove an attacker's fingerprints. They insert a new narrative that directs investigators towards a conclusion that is actually the opposite of the truth. What the ITA 2000's Section 66C identifies as identity theft is uncomfortably similar to this. When an attacker's actual digital fingerprints are systematically removed and replaced by fake ones [7], the act is not just a technicality. No it is an identity theft via evidence itself. The chain of custody which is essential for drawing a clear line between a crime and its perpetrator is broken at this very point. When that thread is snapped the evidence built around it does not weaken. It becomes something that no honest prosecutor will stand behind.

C. Forensic Counter-Analysis as a Constitutional Necessity

As has been established in [2], digital forensics should not only go beyond the simple act of collection but should proceed to Counter-Analysis. This is a procedural requirement to ensure Personal Dignity and Autonomy from state overreach or from criminal forgery. If a court accepts digital evidence without challenging the integrity of its metadata in light of the Anti-Forensic challenges raised in [7], it risks accepting forged evidence. The shift from watermarking protection in 2008 [1] to the current computational methods in 2024 [8] has shown that the legal definition of Due Diligence has to include a mandatory check for hidden metadata.

D. The New Standard of Care: Deep Learning and GDPR

With the advent of Deep Learning models in 2017 [4] and 2022 [6], the Forensic Standard of Care is now different. Manual heuristic checking is no longer "appropriate technical measures" as prescribed by GDPR Article 32, Integrity and Confidentiality. Rather, the automated detection of stegano containers using SRNet [6] and CNN-based steganalysis [4] provides the high probabilistic certainty required to satisfy global data integrity standards. For the report to be considered authoritative in any current litigation scenario it is required to prove the usage of these sophisticated computational techniques [8] to ensure the data is not subject to the sophisticated hiding techniques discussed in [5].

E. Section 85B and the Presumption of Secure Records

According to Section 85B of the ITA 2000 there is a legal presumption that a secure electronic record has not been altered since a specific point in time to which the "secure" status relates. However from the Anti-Forensic overview presented in [7], it is clear that it is possible for the attacker to bypass these security levels by targeting the forensic tools. If the Artifact Wipe or Trail Obfuscation [7] techniques are applied to the Secure record then the legal presumption of integrity is mathematically negated. There is thus a need for a paradigm shift in the way that the Secure Record is defined. Rather than the "Secure Record" being defined by its encryption status it should be defined by its resistance to the advanced data hiding techniques in PDF and Office documents as presented in [5] and [35].

F. The "Attribution Gap" and Section 66C

The provision of Section 66C of the ITA 2000 relates to the punishment for identity theft. From a technical perspective this relates to the Attribution Gap arising from metaheuristic-driven forgery. As explained in [2] and [7], anti-forensic software allows the perpetrator to overwrite Slack Space and Unallocated Clusters with forged information that corresponds to the profile of a legitimate user. By modifying the Manufacturer and Model information in EXIF data [1] or the Author metadata in Microsoft Office files [35], the attacker can by legal means incriminate an unwitting party. The technical possibility of this forces the Forensic Counter-Analysis framework [2] to ensure that digital identity remains traceable and authentic in accordance with the law.

G. Admissibility of AI-Generated Evidence (Section 45 IEA)

As the forensic standards shift towards Deep Learning models [4][6], a new legal dilemma arises in Section 45 of the Indian Evidence Act – Opinion of Experts. In order for the AI-based steganalysis report to be admitted into evidence, the expert has to explain the Probabilistic Characteristics of the model. The SRNet architecture discussed in [6] offers a quantified confidence score for detection. This mathematical certainty is essential in order to satisfy the beyond a reasonable doubt criterion. The shift from manual LSB techniques in 2014 [3] to automated CNN techniques in 2022 [6] is the advancement of the Expert Opinion from subjective observation to objective calculation.

H. Blockchain as the "Standard of Care" under Section 43A)

The body corporate is required to adopt reasonable security practices" in accordance with Section 43A of the ITA 2000. In light of the research by Mahajan and Pandit in 2024 [8], it is

submitted that the adoption of Blockchain and IoT-based integrity systems has redefined the Standard of Care. The hashing mechanism has inherent weaknesses in Length Extension Attacks and Collision Attacks, as discussed in the general literature. By adopting the decentralized computational techniques discussed in [8], a continuous chain of custody can be ensured, which is not vulnerable to Artifact Wiping discussed in [7]. Failure to adopt such decentralized integrity techniques could amount to negligence in accordance with Section 43A.

VII. OPEN RESEARCH CHALLENGES

Despite major breakthroughs in deep learning-based detection and integrity systems based on blockchain technology significant gaps are seen in the field of digital forensics. Deep learning based detection systems have significantly improved the detection of manipulated content whereas blockchain based integrity systems have provided stronger guarantees of data integrity. However these systems have not fully addressed the complex challenges of changing scenarios in digital forensics.

One of the major challenges is the gap between technological and legal requirements. In a legal framework evidence is not only required to be correct but also to be explained and justified. The legal process requires transparency but deep learning-based systems are considered black boxes and are difficult to interpret. The unexplainable nature of these systems makes them less reliable in a legal process despite their technical correctness.

This is an important point because it illustrates a general problem. The development of technology and law has been largely independent of each other. While technology is certainly improving in the field of digital forensics it is not necessarily legal. The effectiveness of a tool is limited by the fact that it cannot be justified in a courtroom.

In order to overcome this problem the design of forensic tools needs to be reconsidered. Not only do transparency and interpretability need to be considered but so do ethical concerns such as data privacy and fairness.

In order for any progress to be made in the future it is important that experts from different fields work together. This is the only way in which digital forensic tools will be able to gain the trust of lawyers.

A. Explainable AI (XAI) for Forensic Testimony

While detection accuracy in and of itself is not particularly informative there is a certain level of performance benchmarking that has been achieved by models like SRNet in [6] and the larger family of CNN architectures in [4]. The problem is that there is a nagging problem lurking just below these performance results a problem that becomes impossible to ignore when any of these is expected to hold up in a courtroom or formal investigative setting proceeding. These models don't explain themselves. They reach conclusions through means that even for their creators can't be understood in any meaningful sense, and the distance between the model concluded this and here's why the model concluded this is vast. This might not be a problem in situations in which the outcome can be validated in another way or in which the risk of being incorrect is low. Forensic testimony is neither of those. A defense attorney does not have to prove an algorithm incorrect he merely has to prove that he can't understand why the algorithm concluded what it did. In a courtroom a black box is not merely a problem it's a risk.

In a legal courtroom according to the Indian Evidence Act (Section 45) an expert is required to explain the basis of his or her opinion to the judge. The major research problem is to design steganalysis models that are able to offer Forensic Explainability. This includes creating structural Heat Maps or saliency maps that identify exactly where and how the PDF's XRef table is being manipulated [5] or which exact pixels are being manipulated in an image by LSB substitution [3]. The gap between the Probabilistic Detection of 2022 models [6] and the Judicial Oversight needed for personal dignity [2] is of critical importance for AI-driven evidence to pass cross-examination.

B. Real-Time Cloud-Edge Steganalysis

As the nature of digital evidence is shifting towards IoT devices and moving directly to Cloud environments [8], the traditional approach of Digital Forensics as a mere post-mortem activity is no longer tenable. The existing state of the art architectures such as SRNet [6] are computationally intensive and demand high end GPU clusters. There is an immediate research imperative for the development of lightweight architectures such as SqueezeNet variants or MobileNet variants that can support on the fly detection at the Edge. If the issue of metadata forgery can be detected during the time of capture (e.g. within the mobile device of a police officer or within the CCTV system) then the Chain of Custody is not compromised before the data reaches the central repository. This is an immediate solution to the existing gap in terms of Expertise and Equipment that is still a bottleneck in the current state of the art [8].

C.Cross-Carrier Universal Steganalysis and Semantic Awareness

All current models can be considered strictly Target-Specific. A model can be optimized to detect LSB-based steganography in JPEG images [3], yet it is nearly useless when applied to structural steganography in PDF metadata [5] or the complex XML-based steganography used in Microsoft Office documents [35]. As your uploaded documents have identified the biggest single barrier facing steganalytic research is the Universal Steganalysis Gap [4]. Ultimately the next steps in steganalytic research will involve the development of Semantic Steganalysis models which do not merely rely on the detection of bit flips but which actually understand the structural logic behind different file formats. Development of the Unified Feature Space required to identify the signatures

of steganographic techniques in many different carriers is the next great frontier

D. Mitigation of Anti-Forensic Sabotage and Self-Healing Tools

The 2024 overview of anti-forensics [7] shows a rather alarming pattern not only are attackers concealing data but they are also actively undermining the forensic process. Attackers are using tools to poison the memory of a forensic workstation or to wipe specific artifacts [7] prior to an investigator even being able to capture them. An important area of research that must be pursued is the creation of Self-Healing Forensic Software. There is something to consider in the notion that a forensic tool, in some manner can be used against itself. If the environment in which the evidence is processed has been secretly undermined then even the most advanced detection techniques have been working under a cloud of assumption. One method to counter this particular weakness is to stop considering integrity as an afterthought in the processing of evidence and to start considering it as an integral part of the tool itself.maintains about itself. Plugging Mahajan and Pandit's [8] computational integrity techniques and blockchain based logging directly into the core of the forensic tool not on top of it but within the core of the tool changes everything. The question is no longer whether the evidence has been altered but whether the device used to investigate has been altered. That's a very different problem to solve and calling it environment protection versus evidence preservation is not just semantics. It changes what you build and where you place your trust.

E.Hardware-Anchored Forensic Authentication and Silicon-Level Trust

From a historical point of view the method proposed in 2008 which relied on EXIF metadata and used BCH error control codes marked a real advance a serious effort to relate integrity to something more akin to hardware. The flaw which became more and more obvious over time was that in the end software remains software, and software can be altered by anybody who has the right access and the right motives. Relating integrity to that level ultimately created a flaw that any determined adversary could exploit.

What the next generation of research in this area is attempting to get to however is slightly more difficult to reach literally. Physical Unclonable Functions and Trusted Execution Environments while not necessarily new concepts in and of themselves taking them and applying them to camera sensors and storage controllers for the purposes of forensics creates a new category of guarantees. The logic while not necessarily easy to perform works like this if the metadata can be secured in the precise moment of creation prior to ever being processed by a software stack that can be altered then the chain of custody starts in the silicon rather than the filesystem. It would have to be altered at a level that, for most adversaries is effectively out of reach, and in many cases even detectable.

This silicon-based level of trust would render the Metadata Forgery discussed in [1] and [2] physically impossible, since any changes would destroy the hardware-based cryptographic bond.

F. Legislative Integration of Algorithmic Evidence

As the Standard of Care in digital forensics continues to evolve from manual heuristic audit trails to AI-driven residual steganalysis techniques [4], [6], the legal system must also adapt. One of the major non-technical challenges is the Standardization Gap the lack of a legal protocol for the introduction of blockchain-based evidence logs [8] or AI-driven stegodetection reports. Further research is needed to develop a Digital Evidence Standard that acknowledges Probabilistic Certainty as a legitimate legal principle so that the advanced defenses discussed in this survey [1]–[8] are not only technically revolutionary, but also legally paradigm-shifting

VIII. CONCLUSION

The move from manual heuristic testing to deep learning as the new standard in forensic practice from EXIF marking and BCH error control codes in the past to the sophisticated residual steganalysis of SRNet in the latest standard is not merely an evolution but a revolution in the definition of digital reliability. The evolution of the defense of digital evidence which this survey has systematically tracked over nearly two decades has highlighted the evolution of the Defensive position of Theme 12 in response to the sophisticated Offensive metadata manipulation and data hiding techniques.

As has been pointed out in the course of the survey the Verification Paradox [2] is known to present the very core of the problem for the contemporary jurisprudence. A file that looks immaculate from a visual perspective and bears a valid cryptographic hash can contain harmful changes in its LSB layers [3], PDF XRef tables [5], or Microsoft Office XML metadata [35]. This makes for a Forensic Blind Spot that challenges the very legal framework of Section 65B of the ITA 2000. Without the development of a Counter Analysis framework [2], the legal system is in peril of accepting poisoned evidence that has been subjected to anti-forensic artifact wiping and trail obfuscation techniques as outlined in [7].

Furthermore with the advent of Convolutional Neural Networks (CNNs) in 2017 [4] and optimized deep learning models on 2022 [6], the Forensic Standard of Care has been raised to a level where human audit is no longer feasible. Authentic and Unaltered under the Indian Evidence Act will require a digital record to pass a rigorous computational audit. The move to blockchain based integrity and IoT based computational solutions [8], representing the final frontier in establishing a continuous chain of custody that is mathematically immune to Attribution Gap and Identity Theft issues raised in Section 66C.

Finally the five open research challenges indicate the need to have an Explainable AI to support forensic evidence for the

development of hardware anchored silicon based trust mechanisms for a critical multi-disciplinary research agenda for the coming decade. The five open research challenges cut across different domains including artificial intelligence, cybersecurity, digital forensics and hardware security.

The closing of these gaps is not optional it is essential for the integrity of the modern judicial and investigative system. So long as the Detection Forgery Gap remains sophisticated adversaries will continue to take advantage of weaknesses in current technology to produce convincing forgeries, deceive legal processes and subvert the rule of law. The Detection Forgery Gap is the difference between the rapid advancement of forgery techniques and the slower pace of detection techniques.

In order to resolve this problem, it is mandatory that advanced deep learning based detection mechanisms are integrated with strong integrity solutions. Deep learning based solutions can help in improving the accuracy of the detection of tampered digital information. At the same time decentralized solutions such as the blockchain based solution can help in maintaining the authenticity of digital information.

With a combination of these approaches the field of digital forensics can take a step closer to realizing its founding promise determining the truth of a digital record with a high level of mathematical assurance and legal acceptability.

This addition has a vital role in building trust in digital evidence and protecting the justice system from new technological threat.

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